

Photoelectric effect

Let $\psi_a(\mathbf{r})$ be the ground state for hydrogen.

$$\psi_a(\mathbf{r}) = \frac{1}{\sqrt{\pi a^3}} \exp(-r/a)$$

Let $\psi_b(\mathbf{r})$ be the wave function for a free electron.

$$\psi_b(\mathbf{r}) = \frac{1}{\sqrt{l^3}} \exp(i\mathbf{k} \cdot \mathbf{r}) = \frac{1}{\sqrt{l^3}} \exp(ikr \cos \theta)$$

Find the transition rate $R_{a \rightarrow b}$ using Fermi's golden rule

$$R_{a \rightarrow b} = \frac{2\pi}{\hbar} |\langle \psi_b | H_1 | \psi_a \rangle|^2 \rho(E_b)$$

§1

Start by finding the perturbing Hamiltonian H_1 .

We have for a particle with charge q

$$H = \frac{1}{2m} (\mathbf{p} - q\mathbf{A}) \cdot (\mathbf{p} - q\mathbf{A})$$

Expand the inner product.

$$H = \frac{1}{2m} (\mathbf{p}^2 - \mathbf{p} \cdot q\mathbf{A} - q\mathbf{A} \cdot \mathbf{p} + q^2 \mathbf{A}^2)$$

Hence the first-order perturbation is

$$H_1 = -\frac{q}{2m} (\mathbf{p} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p})$$

For an electron we have $q = -e$.

$$H_1 = \frac{e}{2m} (\mathbf{p} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p})$$

For any wave function $\psi(\mathbf{r}, t)$

$$(\mathbf{p} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p})\psi(\mathbf{r}, t) = 2\mathbf{A} \cdot \mathbf{p}\psi(\mathbf{r}, t) - i\hbar\psi(\mathbf{r}, t)\nabla \cdot \mathbf{A} \quad (1)$$

Hence for the Coulomb gauge $\nabla \cdot \mathbf{A} = 0$ we have the identity

$$\mathbf{p} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p} = 2\mathbf{A} \cdot \mathbf{p}$$

Hence

$$H_1 = \frac{e}{m} \mathbf{A} \cdot \mathbf{p}$$

§2

Let $\mathbf{E}$ be the electric field

$$\mathbf{E} = E_0 \sin(\mathbf{k} \cdot \mathbf{r} - \omega t) \boldsymbol{\epsilon}$$

where $\boldsymbol{\epsilon}$ is a polarization vector such that $|\boldsymbol{\epsilon}| = 1$ and $\mathbf{k} \cdot \boldsymbol{\epsilon} = 0$.

Then the magnetic vector potential $\mathbf{A}$ is

$$\mathbf{A} = -\frac{E_0}{\omega} \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \boldsymbol{\epsilon}$$

so that in SI units

$$\mathbf{E} = -\frac{\partial}{\partial t} \mathbf{A} \quad (2)$$

Hence

$$H_1 = \frac{e}{m} \mathbf{A} \cdot \mathbf{p} = -\frac{eE_0}{m\omega} \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \boldsymbol{\epsilon} \cdot \mathbf{p}$$

In exponential form

$$H_1 = -\frac{eE_0}{2m\omega} \exp(i\mathbf{k} \cdot \mathbf{r} - i\omega t) \boldsymbol{\epsilon} \cdot \mathbf{p} - \frac{eE_0}{2m\omega} \exp(-i\mathbf{k} \cdot \mathbf{r} + i\omega t) \boldsymbol{\epsilon} \cdot \mathbf{p}$$

Use the dipole approximation $\exp(i\mathbf{k} \cdot \mathbf{r}) \approx 1$.

$$H_1 = -\frac{eE_0}{2m\omega} \exp(-i\omega t) \boldsymbol{\epsilon} \cdot \mathbf{p} - \frac{eE_0}{2m\omega} \exp(i\omega t) \boldsymbol{\epsilon} \cdot \mathbf{p}$$

Go back to cosine form.

$$H_1 = -\frac{eE_0}{m\omega} \cos(\omega t) \boldsymbol{\epsilon} \cdot \mathbf{p}$$

By the identity

$$\mathbf{p} = \frac{im}{\hbar} (H_0 \mathbf{r} - \mathbf{r} H_0)$$

we have

$$\langle \psi_b | \mathbf{p} | \psi_a \rangle = \frac{im}{\hbar} (E_b - E_a) \langle \psi_b | \mathbf{r} | \psi_a \rangle$$

Hence

$$H_1 = -ieE_0 \frac{\omega_0}{\omega} \cos(\omega t) \boldsymbol{\epsilon} \cdot \mathbf{r}$$

where

$$\omega_0 = \frac{E_b - E_a}{\hbar}$$

For the photoelectric effect we have $\omega_0 = \omega$ hence

$$H_1 = -ieE_0 \cos(\omega t) \boldsymbol{\epsilon} \cdot \mathbf{r}$$

Rotate the lab frame so that $\boldsymbol{\epsilon} \cdot \mathbf{r} = z$.

$$H_1 = -ieE_0 \cos(\omega t) z$$

§3

For perturbing Hamiltonians of the form

$$H_1 = V(\mathbf{r}) \cos(\omega t)$$

we have

$$|\langle \psi_b | H_1 | \psi_a \rangle|^2 = |H_{ba}|^2$$

where H_{ba} is the transition amplitude

$$H_{ba} = \frac{1}{2} \int \psi_b^* V(\mathbf{r}) \psi_a d\mathbf{r}$$

Hence for

$$V(\mathbf{r}) = -ieE_0 z$$

we have in polar coordinates

$$H_{ba} = -\frac{ieE_0}{2} \frac{1}{\sqrt{\pi a^3 l^3}} \int_0^\infty \int_0^\pi \int_0^{2\pi} \exp(ikr \cos \theta - r/a) z r^2 \sin \theta dr d\theta d\phi$$

Integrate over ϕ (multiply by 2π).

$$H_{ba} = -i\pi eE_0 \frac{1}{\sqrt{\pi a^3 l^3}} \int_0^\infty \int_0^\pi \exp(ikr \cos \theta - r/a) z r^2 \sin \theta dr d\theta$$

Change of variable $u = \cos \theta$.

$$H_{ba} = -i\pi eE_0 \frac{1}{\sqrt{\pi a^3 l^3}} \int_0^\infty \int_{-1}^1 \exp(ikru - r/a) z r^2 dr du$$

Solve the integral over u .

$$H_{ba} = -i\pi eE_0 \frac{1}{\sqrt{\pi a^3 l^3}} \int_0^\infty \frac{1}{ikr} [\exp(ikr - r/a) - \exp(-ikr - r/a)] z r^2 dr \quad (3)$$

Cancel i and r .

$$H_{ba} = -\pi eE_0 \frac{1}{\sqrt{\pi a^3 l^3}} \frac{1}{k} \int_0^\infty [\exp(ikr - r/a) - \exp(-ikr - r/a)] z r dr$$

Substitute $z = r \cos \theta$. This θ is different from θ in $\mathbf{k} \cdot \mathbf{r} = kr \cos \theta$.

$$H_{ba} = -\pi eE_0 \frac{1}{\sqrt{\pi a^3 l^3}} \frac{\cos \theta}{k} \int_0^\infty [\exp(ikr - r/a) - \exp(-ikr - r/a)] r^2 dr$$

Solve the integral.

$$H_{ba} = -\pi eE_0 \frac{1}{\sqrt{\pi a^3 l^3}} \frac{\cos \theta}{k} \left[-\frac{2}{(ik - 1/a)^3} + \frac{2}{(-ik - 1/a)^3} \right] \quad (4)$$

Rewrite as

$$H_{ba} = -\pi eE_0 \frac{1}{\sqrt{\pi a^3 l^3}} \cos \theta i a^4 \left[\frac{16}{(a^2 k^2 + 1)^3} - \frac{4}{(a^2 k^2 + 1)^2} \right] \quad (5)$$

§4

For the transition density we have

$$|\langle \psi_b | H_1 | \psi_a \rangle|^2 = |H_{ba}|^2 = \pi e^2 E_0^2 \frac{a^5}{l^3} \cos^2 \theta \left[\frac{16}{(a^2 k^2 + 1)^3} - \frac{4}{(a^2 k^2 + 1)^2} \right]^2 \quad (6)$$

Hence by Fermi's golden rule

$$\begin{aligned} R_{a \rightarrow b} &= \frac{2\pi}{\hbar} |\langle \psi_b | H_1 | \psi_a \rangle|^2 \rho(E_b) \\ &= \frac{2\pi^2}{\hbar} e^2 E_0^2 \frac{a^5}{l^3} \cos^2 \theta \left[\frac{16}{(a^2 k^2 + 1)^3} - \frac{4}{(a^2 k^2 + 1)^2} \right]^2 \rho(E_b) \end{aligned}$$

The density of states for a free electron is

$$\rho(E_b) = \left(\frac{l}{2\pi} \right)^3 \frac{\sqrt{2m^3 E_b}}{\hbar^3} = \left(\frac{l}{2\pi} \right)^3 \frac{k}{a\alpha \hbar c} \quad (7)$$

Hence

$$R_{a \rightarrow b} = \frac{ka^4}{4\pi\alpha\hbar^2 c} e^2 E_0^2 \cos^2 \theta \left[\frac{16}{(a^2 k^2 + 1)^3} - \frac{4}{(a^2 k^2 + 1)^2} \right]^2 \quad (8)$$

§5

Verify dimensions.

$$\begin{aligned} \alpha &= [\text{V}] \\ a &= [\text{m}] \\ k &= [\text{m}^{-1}] \\ \omega &= [\text{s}^{-1}] \\ e &= [\text{C}] \\ E_0 &= [\text{N C}^{-1}] = [\text{kg m s}^{-2} \text{C}^{-1}] \\ \hbar &= [\text{J s}] = [\text{kg m}^2 \text{s}^{-1}] \\ \frac{k}{a\alpha\hbar c} &= \frac{[\text{m}^{-1}]}{[\text{m}] [\text{J s}] [\text{m s}^{-1}]} = \frac{1}{[\text{J}] [\text{m}^3]} \\ R_{a \rightarrow b} &= \frac{[\text{m}^{-1}] [\text{m}]^4}{[\text{J s}]^2 [\text{m s}^{-1}]} [\text{N}]^2 = [\text{s}^{-1}] \end{aligned} \quad (9)$$

§6

Eigenmath script